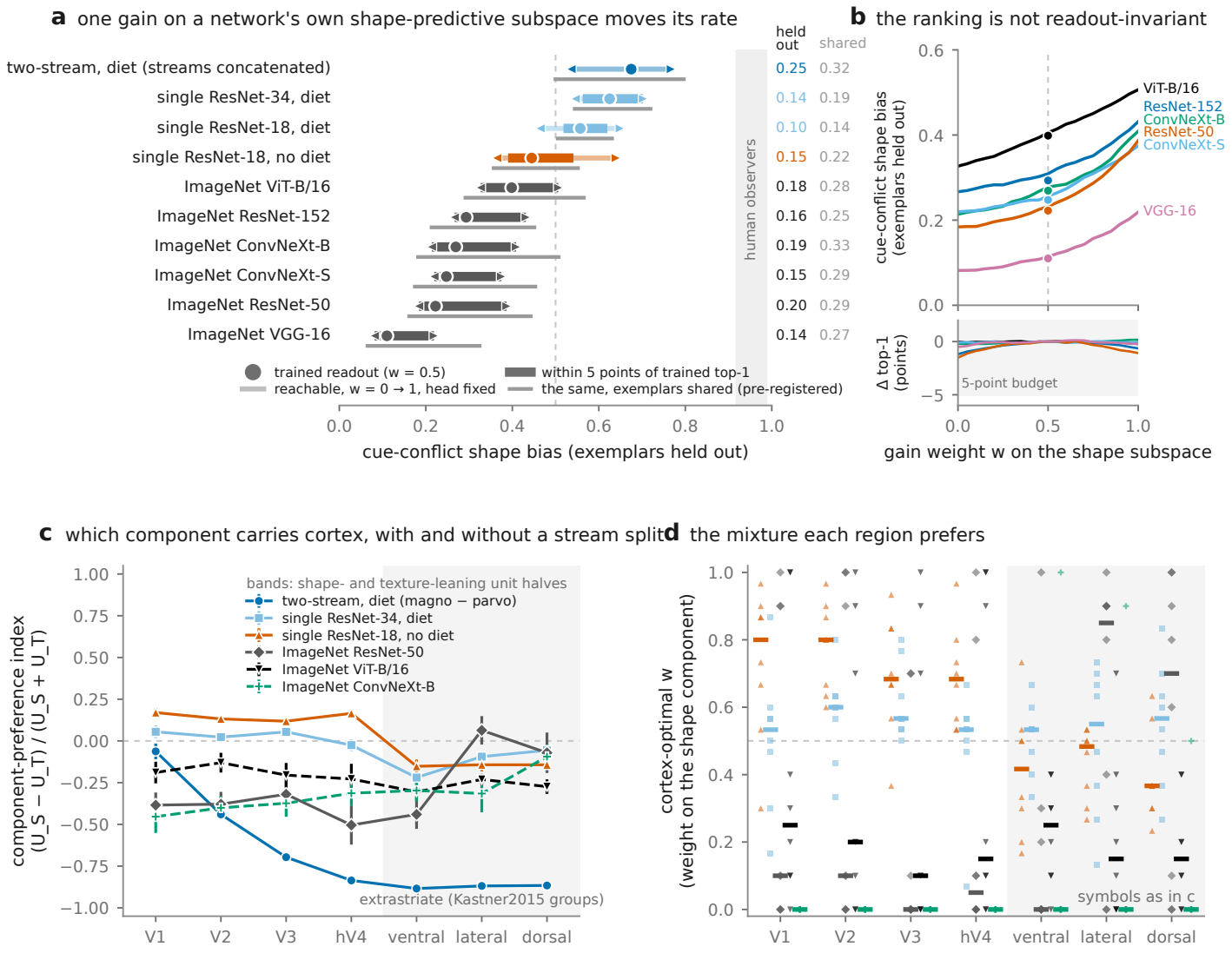
a one gain on a network's own shape-predictive subspace moves its rate
held out
shared
two-stream, diet (streams concatenated) 0.25 0.32
single ResNet-34, diet 0.14 0.19
single ResNet-18, diet 0.10 0.14
single ResNet-18, no diet 0.15 0.22
ImageNet ViT-B/16 0.18 0.28
ImageNet ResNet-152 0.16 0.25
ImageNet ConvNeXt-B 0.19 0.33
ImageNet ConvNeXt-S 0.15 0.29
ImageNet ResNet-50 0.20 0.29
ImageNet VGG-16 0.14 0.27
human observers
trained readout (w = 0.5)
reachable, w = 0 → 1, head fixed
within 5 points of trained top-1
the same, exemplars shared (pre-registered)
0.0 0.2 0.4 0.6 0.8 1.0
cue-conflict shape bias (exemplars held out)
b the ranking is not readout-invariant
cue-conflict shape bias (exemplars held out)
0.0 0.2 0.4 0.6
ViT-B/16
ResNet-152
ConvNeXt-B
ResNet-50
ConvNeXt-S
VGG-16
Δ top-1 (points)
0
−5
5-point budget
0.0 0.5 1.0
gain weight w on the shape subspace
c which component carries cortex, with and without a stream split
component-preference index (U_S − U_T) / (U_S + U_T)
1.00 0.75 0.50 0.25 0.00 −0.25 −0.50 −0.75 −1.00
bands: shape- and texture-leaning unit halves
two-stream, diet (magno − parvo)
single ResNet-34, diet
single ResNet-18, no diet
ImageNet ResNet-50
ImageNet ViT-B/16
ImageNet ConvNeXt-B
extrastriate (Kastner2015 groups)
V1 V2 V3 hV4 ventral lateral dorsal
d the mixture each region prefers
cortex-optimal w (weight on the shape component)
1.0 0.8 0.6 0.4 0.2 0.0
symbols as in c
V1 V2 V3 hV4 ventral lateral dorsal